

Jake van Halder

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Skills

Languages: TypeScript, JavaScript, Golang, C/C++, C#, Python

Tools: React, Vite, Git, Linux, Figma

Experience

Software Engineer Intern, Schweitzer Engineering Laboratories – Pullman, WA April 2025 – Present

- Develop enterprise-grade web applications using React (TypeScript) and Golang, improving performance and maintainability of internal and external tools.
- Built a **Jenkins artifact pipeline** using Shell scripting and Node.js to aggregate SEL device data into static HTML dashboards, reducing data retrieval time.
- Implement end-to-end testing pipelines with Playwright and Vitest, increasing test coverage.
- Collaborate in an Agile team environment, contributing to code reviews and CI/CD workflows.

Research Aid 2 (Frontend Developer), Washington State University – Pullman, WA October 2024 – Present

- Lead frontend development of a GIS-based watershed analytics platform supporting hydrologic research across Washington and Oregon.
- Build interactive geospatial features using React (TypeScript) and Leaflet, enabling real-time watershed visualization and data exploration.
- Integrate frontend components with a Django backend to dynamically render watershed properties and time-series datasets.
- Optimize map rendering and spatial data handling for improved performance with large shapefiles.

Projects

Bite | React Native (Expo), TypeScript, Node.js, PostgreSQL

- Developing a cross-platform mobile application using React Native (Expo) to manage recipes, track food expiration, and monitor grocery spending.
- Designing a Python-based backend with FastAPI to support recipe storage, user authentication, and ML-powered ingredient extraction.
- Prototyping machine learning workflows in PyTorch to classify and extract ingredients from scanned recipe images.
- Structuring a PostgreSQL database schema to support scalable user, recipe, and expiration-tracking models.

CGDJ-BulletHell | C#, MonoGame, Git

- Lead a 4-person team in developing a 2D bullet-hell shooter inspired by Touhou Project using MonoGame.
- Architected the project's modular file structure, separating game actors, combat systems, patterns, and stage logic for scalability and maintainability.
- Managed project workflow using GitHub Issues and pull requests, conducting code reviews and coordinating feature integration.
- Established development standards and task templates to streamline collaboration and maintain consistent code quality.

Education

Washington State University – BS in Computer Science

Expected Graduation: May 2026

Club – Software Development Club

September 2024 - Present